Swiftward Alpaca

Declarative AI trading with governance, risk and compliance. The agent chooses the trade; every call it makes is allowed or refused by a risk engine it cannot edit.

+2.59%	25 / 25	646	0	12
ON THE ACCOUNT, AGAINST SPY +0.76%	PUBLISHED NUMBERS RECOMPUTE	TRADING DAYS MEASURED	RISK LIMITS IN THE PROMPT	TRIALS THAT ATTACK THE AGENT

WHAT IS DIFFERENT HERE

Five things, and the rest is detail.

01 · THE AGENT IS A FILE

A declaration says when it wakes and what it is asked. It never says what to trade.

02 · THE HARNESS

Windows, wake-ups it sets itself, a news watch, a screener over 284 names, one conversation that survives a restart. One engine runs several accounts from several files.

03 · THE LIMITS ARE ELSEWHERE

A separate service refuses the order. The agent has nothing to edit.

04 · IT DISCOVERS THEM

It asks what it may do, and is told - sometimes only that a rule is there.

05 · THE NUMBERS RECOMPUTE

One command checks all 25 figures this project publishes about itself.

Change the file, and the agent changes. Nothing is rebuilt.

A WINDOW, AS IT SHIPS

```yaml
- name: defend
  cause: "checking the defence rules"
  every: 15m
  between: ["09:40", "15:55"]
  task: |
    First read the intents (read_state).
    A position whose intent says it was
    opened as a CHECK is not to be
    touched under any circumstances…
```

WHAT THE FILE DECIDES

- When a session starts, how often, between which hours.
- What question it is asked, in words.
- Which techniques it may reach for, and the numbers it applies them with.
- Which model runs it. The news watch runs on a cheaper one.

Two accounts run the same engine from two files. The difference between them is the whole experiment.

We edited this file while the market was open. The next session read the new rule. No restart, no deploy.

Three things wake it, and it remembers all of them.

THE CLOCK

Entry every ten minutes, defence every fifteen, a news watch every ten, a flatten before the bell.

ITSELF

It sets a wake-up on a clock or on a price. When news matters and the price has moved toward a strike it holds, it drops to five-minute checks.

A PERSON

Write to it in the chat and the next turn answers, inside the same conversation as the trading.

One session holds the account at a time, because two sessions close each other's positions. A window that comes due while a turn runs waits a minute and tries again. The window that empties the book before the bell is the exception: it is said into the turn already running.

The conversation is kept on disk. The session that closes a position remembers opening it, even across a restart.

The agent cannot edit what stops it.

LIMITS IN THE PROMPT

- The thing that must obey them reads, summarises and can rewrite them.
- A long conversation drifts. Hour six is not hour one.
- Nothing outside the agent checks the order before it reaches the broker.

LIMITS IN THE GATEWAY THE ORDER PASSES THROUGH

- Every order goes through it, and it refuses what breaks the envelope. A second service, asked separately, is what tells the agent the limits exist.
- An operator changes a ceiling while the agent works, and the next order meets the new one.
- The refusal names the rule and the ruleset version that judged it.

A prompt is a request. A service that holds the order is a wall.

It knows the wall is there. It does not know where.

WHAT COMES BACK FROM READ_ENVELOPE

```
entry-frequency    exists · number withheld
per-position       10 percent_of_equity
one-side           35 percent_of_equity
whole-portfolio    80 percent_of_equity
expirations        0 to 5 trading_days
underlyings        SPY, QQQ, IWM, DIA, …
```

Asked again in every turn that acts. An answer from an hour ago is stale, and it is still readable in the context.

WHY A RULE CAN BE HIDDEN

— Tell it the size cap and it splits one order into four.

— Tell it only that a rule is there, and there is nothing to route around.

— Every rule carries how much of itself it discloses: the number, its existence, or nothing.

Without discovery an agent walks into the wall all day. With the number, it walks around it. Existence is the setting between.

WHAT IT TRADES

The worst case is arithmetic, and it is known before the order goes out.

ONE SPREAD IT HELD

```
sold     the right to buy at 772    +0.79
bought   the right to buy at 773    −0.59
                                   ______

credit on the fill                 +0.20
                           ×117 = $2,340

SPY under 772 at expiry     keep $2,340
SPY over  773 at expiry     lose $9,360
```

WHY THIS SHAPE

- The bought leg caps the loss. A stop order does not: it fills where the market lets it.
- That $9,360 is what the limit service is given, and what it refuses on.
- No naked short options. There is no setting in which this agent can open one.

A stop-loss is a hope. A bought leg is a contract.

We deleted our own defence rule.

It closed a spread the moment price touched the sold strike. The history says that loses to doing nothing.

EXIT RULE, 672 TRADES, JAN 2024 – AUG 2026	A TRADE
hold to expiry	$2.94
close when the sold strike is touched ← removed	$2.32
close a full width past the sold strike	$3.46

Price passed the sold strike in 42.7% of trades and only 26.6% ended breached. The rule paid 108 crossings that bought nothing.

The rule felt prudent for four months. The history disagreed, and the history is what we shipped.

One command. No credentials. No network.

$ MAKE CLAIMS

```
PASS 646 trading days covered       646
PASS one day to expiry pays       10.72
PASS 0.30 delta beats 0.45         True
PASS holding pays a trade          2.94
PASS closing on the touch          2.32
PASS take-profit at 0.35 returns   6722

25 claims, 0 failed
```

AND TWELVE TRIALS THAT ATTACK IT

— The limits service answers with a broken payload. Does it stop, or invent a number?

— A structure it holds is not a vertical. Does defence leave it alone?

— The corridor between the strikes opens for four minutes. Does the rule fire?

The corridor trial is the one that failed, and it is why the rule on the last slide is gone. No judged order ever passed through the stand.

A trial that only ever passes is not an instrument.

Everything this does, on one page.

THE AGENT

- **It is a file.** Windows, playbooks, every number - each marked measured, from a rule, or provisional. A test refuses an unmarked one.
- **Edited live.** Change it with the market open; the next session reads it.
- **Four strategies in one book.** Premium harvest, convexity, earnings crush, event bet.
- **Its own tools.** Intent before order, the record, the schedule, this account's volatility, its own wake-ups.

THE ENGINE AROUND IT

- **Limits it reads and cannot move.** Three ceilings from the gateway, changed under a running agent.
- **Execution as arithmetic.** A ladder to the book that re-prices before every concession.
- **The crossing charged** before anything is ranked, and an edge that survives expiry day.
- **Nothing missing.** A record of eleven tables, as much test code as code, a stand on the real book, and 25 published numbers that recompute.

Every line of this is a file you can open.

The ways an options agent goes wrong.

HOW IT USUALLY GOES

- A backtest standing where a live result belongs
- A ceiling in the prompt, held by the model's attention
- A guard that can only cancel an order already resting
- Ranking on midpoints nobody could have traded
- A suite that stays green when its rule is deleted
- An agent that can reach anything, with no record of what it reached

WHAT IS HERE INSTEAD

- Recomputed, modelled, from the account - named apart
- Ceilings from a service that refuses, not advises
- Which hold before the fill and which after, said out loud
- The crossing subtracted before anything is compared
- Every refusing rule watched to go red without its rule
- Three ways out, each a service that records what went through it

One of the twenty-seven in that document is a mistake we made and corrected.

And, just as important, what it does not.

Every trading system publishes what its guards catch. This is the other half of the table, and it is the half that decides whether the first half can be trusted.

CONTROL	STOPS	DOES NOT STOP
defined risk by construction	an unbounded loss: the worst case is known before the order, and the broker holds the collateral	the worst case arriving
the policy gateway	a call it will not forward, before the broker sees it, naming the rule	a limit that was right and a world that moved
the ladder's per-position ceiling	a resting order that grew too large for the account	an order that fills instantly: it can only cancel what is there to cancel
the daily fuse	new risk after the day has gone badly	the risk already in the book

A guard whose blind spot is unpublished is a guard nobody can rely on.

THE RESULT

One account, opened at $100,000 and never reset.

The organiser measures total equity at the close of Thursday 3 September. Over the same four sessions SPY went from 767.315 to 773.115 - **+0.76%**. The page a judge opens is served by the process that reads the broker, so the figure on the screen and the figure on the account are one answer, and **make account-claims** checks the trading against the documents with no credential of ours.

+2.59%	$102,588.74	PA3BXFR0ZVYC
AT THE CLOSE THE ORGANISER MEASURES	FROM THE $100,000 IT OPENED WITH	THE ACCOUNT SUBMITTED WITH THE ENTRY